

MALNAD COLLEGE OF ENGINEERING

HASSAN-573202

(An Autonomous Institution under VTU , Belagavi)

Hassan, Karnataka 573202, India



Activity report on:

“Introduction to AI and it’s use cases”

(ETC525O)

SUBMITTED BY:

ROOPA	4MC25CS246
SHRAVYA M M	4MC25CS283
SINCHANA	4MC25CS289
SHOBITHA	4MC25CS345

UNDER THE GUIDANCE OF :

Mr . Ravi kumar

Assistant Professor

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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INTRODUCTION:

Online Exam Cheating Detection Using Classification (SVM)

Introduction

With the rapid growth of online education and remote examinations, maintaining academic honesty has become a major challenge. In physical classrooms, invigilators monitor students directly, but in online exams, students attend tests from different locations using computers or mobile devices. This creates opportunities for cheating such as tab switching, searching answers online, communicating with others, or using unfair tools.

To solve this problem, machine learning techniques can be used to detect suspicious behavior automatically. One effective approach is **Classification using Support Vector Machine (SVM)**. The system analyzes student behavior data and classifies it into:

- **Normal Behavior**
- **Suspicious Behavior**

This helps educational institutions maintain fairness and integrity during online examinations.

PROBLEM STATEMENT:

With the rise of online examinations, maintaining academic integrity has become challenging. The system logs data such as time spent per question, frequency of the tab switching, mouse movements patterns, and answer submission timing. We should build a code to classify whether a student's behavior is "Normal" or "Suspicious," helping to detect potential cheating.

Objectives of the System

The main objectives are:

1. Detect cheating behavior during online exams.
2. Monitor student activities automatically.
3. Reduce manual invigilation effort.
4. Improve fairness in examinations.
5. Use machine learning for intelligent decision-making.

INFORMATION OF THE CLASSIFICATION AND SVM:

Data Collected by the System

The system continuously records various student activities during the examination.

Important Features Collected

Feature	Description
Time spent per question	Measures how long a student answers each question
Tab switching frequency	Counts how many times the student leaves the exam window
Mouse movement patterns	Detects abnormal mouse behavior
Keyboard activity	Monitors typing speed and unusual patterns
Answer submission timing	Detects extremely fast or delayed submissions
Webcam monitoring	Optional feature to track face direction
Copy-paste attempts	Detects unauthorized pasting
Browser activity	Checks suspicious browser usage

What is Classification?

Classification is a machine learning technique where the computer predicts the category of data.

In this project:

- Input → Student behavior data
- Output → Normal or Suspicious

This is called **binary classification** because there are only two classes.

What is SVM (Support Vector Machine)?

Support Vector Machine (SVM) is a supervised machine learning algorithm used for classification problems.

It works by finding the best boundary (called a hyperplane) that separates different classes of data. In this system:

- One side → Normal students
- Other side → Suspicious students

METHODOLOGY:

Working Principle of SVM:

The SVM algorithm:

1. Takes training data.
2. Finds patterns in student behavior.
3. Creates a decision boundary.
4. Classifies new student activities.

For example:

- Low tab switching + normal timing → Normal
- High tab switching + unusual timing → Suspicious

Mathematical Representation of SVM:

The decision boundary equation of SVM is:

$$w \cdot x + b = 0$$

Where:

- w = weight vector
- x = input features
- b = bias

Classification rule:

- If result > 0 → Normal
- If result < 0 → Suspicious

SYSTEM ARCHITECTURE:

Step-by-Step Process

1. Data Collection

The system collects behavioral data from students during exams.

2. Data Preprocessing

The collected data is cleaned and normalized.

Example:

- Removing missing values
- Scaling numerical values

3. Feature Extraction

Important features like tab switches and timing are selected.

4. Model Training

The SVM model is trained using previous labeled data.

5. Classification

The trained model predicts whether behavior is suspicious or normal.

6. Alert Generation

If suspicious activity is detected, the system alerts administrators.

EXAMPLE DATASET:

Student	Tab Switches	Time per Question	Mouse Activity	Result
A	1	Normal	Stable	Normal
B	10	Very Fast	Random	Suspicious
C	0	Normal	Stable	Normal
D	15	Irregular	Abnormal	Suspicious

ADVANTAGES OF SVM IN THIS PROJECT:

Advantage	Explanation
High accuracy	Works well for classification
Effective with small datasets	Gives good results even with limited data
Handles complex patterns	Detects cheating behaviors effectively
Memory efficient	Uses support vectors only
Reliable performance	Suitable for online monitoring

Challenges in the System:

1. Privacy Issues

Monitoring webcam and browser activity may affect student privacy.

2. False Detection

Sometimes honest students may appear suspicious.

3. Internet Problems

Poor connectivity may affect data collection.

4. Hardware Differences

Different devices may behave differently.

Technologies Used:

<u>Technology</u>	<u>Purpose</u>
Python	Programming language
Scikit-learn	Machine learning library
OpenCV	Webcam monitoring
Pandas	Data handling

Technology

Purpose

NumPy	Numerical computation
Flask/Django	Web application backend

Evaluation Metrics:

The system performance is measured using:

Metric

Meaning

Accuracy	Overall correctness
Precision	Correct suspicious detections
Recall	Ability to detect cheating
F1-score	Balance between precision and recall

Accuracy formula:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Where:

- TP = True Positive
- TN = True Negative
- FP = False Positive
- FN = False Negative

RESULTS:

Applications:

This system can be used in:

- Online school examinations
- University entrance tests
- Certification exams
- Government recruitment exams
- Corporate training assessments

Future Improvements:

Future versions may include:

1. AI-based face tracking
2. Voice detection
3. Eye movement analysis
4. Deep learning models
5. Real-time dashboard monitoring
6. Mobile exam security systems

CONCLUSION:

Online Exam Cheating Detection using Classification (SVM) is an intelligent solution for maintaining academic integrity in digital examinations. By analyzing student behavior patterns such as tab switching, answer timing, and mouse movement, the system can effectively identify suspicious activities.

Support Vector Machine (SVM) provides high accuracy and reliable classification, making it suitable for educational institutions conducting online examinations. Although there are challenges such as privacy concerns and false detections, future advancements in artificial intelligence can further improve the efficiency and reliability of the system.

This technology plays an important role in creating fair, transparent, and secure online examination environments.

CODE USED:

ONLINE EXAM CHEATING DETECTION USING SVM

MULTIPLE STUDENTS INPUT AND OUTPUT

from sklearn import svm

TRAINING DATASET

```
X = [  
    [45, 0, 120, 30], [50, 1, 100, 25], [48, 0, 110, 28], [20, 8, 15, 5], [15, 10, 10, 3], [18, 7, 20, 4], [55, 1, 130, 35], [52, 2, 125, 30],  
    [17, 9, 18, 4], [49, 0, 115, 27]  
]
```

LABELS

0 = NORMAL

1 = SUSPICIOUS

```
y = [0, 0, 0, 1, 1, 1, 0, 0, 1, 0]
```

CREATE AND TRAIN MODEL

```
model = svm.SVC(kernel='linear')
```

```
model.fit(X, y)
```

NUMBER OF STUDENTS

```
n = int(input("Enter number of students: "))
```

INPUT AND OUTPUT

```
for i in range(n):
```

```
    print("\nENTER DETAILS OF STUDENT", i + 1)
```

```
    time_per_question = int(input("Average time per question: "))
```

```
    tab_switches = int(input("Number of tab switches: "))
```

```
    mouse_movements = int(input("Mouse movements count: "))
```

```
    submission_time = int(input("Answer submission time: "))
```

STUDENT DATA

```
student = [[  
    time_per_question,  
    tab_switches,  
    mouse_movements,  
    submission_time  
]]
```

PREDICTION

```
result = model.predict(student)
```

OUTPUT

```
if result[0] == 0:
```

```
    print("Behavior Status: NORMAL")
```

```
else:
```

```
    print("Behavior Status: SUSPICIOUS")
```

OUTPUT:

Enter number of students: 4

ENTER DETAILS OF STUDENT 1

Average time per question: 50

Number of tab switches: 1

Mouse movements count: 120

Answer submission time: 30

Behavior Status: NORMAL

ENTER DETAILS OF STUDENT 2

Average time per question: 15

Number of tab switches: 10

Mouse movements count: 10

Answer submission time: 3

Behaviour Status: SUSPICIOUS

ENTER DETAILS OF STUDENT 3

Average time per question: 34

Number of tab switches: 15

Mouse movements count: 20

Answer submission time: 6

Behavior Status: SUSPICIOUS

ENTER DETAILS OF STUDENT 4

Average time per question: 5

Number of tab switches: 54

Mouse movements count: 45

Answer submission time: 20

Behavior Status: SUSPICIOUS

*****Thank you*****